MA1522 LINEAR ALGEBRA FOR COMPUTING
National University of Singapore

Exercise Set F (Final Exam Revision)
Name: _____
Date: _____

Linear Systems, Subspaces, Orthogonality, and Eigenvalues

Lecture slides Chapters 1–7 • Lessons 1–23 • Tutorials 1–11

HOW TO USE THIS SET

The final examination on 24 November 2026 is a two-hour, closed-book paper with one A4 helpsheet and an approved calculator. Section A has 16 multiple-choice questions of 2 marks each; Section B has 5 written questions worth 68 marks, for which full working must be shown. Parts A to F of this set revise the seven chapters in order, with most of the weight on written questions. Part G is a complete mock paper in the examination format, to be sat in one sitting of two hours.

Every problem here is new: none is copied from the slides, the tutorials, the quizzes, or the past papers, although each is modelled on a question type that appears there. Every written problem is followed by a ruled panel sized to the work it should take; multiple-choice questions have none. Answers to all problems begin on the last pages; consult them only after a complete attempt.

Distribution of Problems

Part	Topic	Chapters	Problems
A	Linear Systems, Matrices, and Determinants	1–2	1–10
B	Euclidean Vector Spaces	3	11–18
C	Subspaces Associated to a Matrix and Rank	4	19–24
D	Orthogonality and Least Squares	5	25–33
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Notation

Vectors are column vectors and are written in bold, as are matrices. $\boldsymbol{A}^{\mathrm{T}}$ is the transpose, $\mathrm{Col}(\boldsymbol{A})$, $\mathrm{Row}(\boldsymbol{A})$ and $\mathrm{Null}(\boldsymbol{A})$ are the column space, row space and nullspace of $\boldsymbol{A}$, and $[\boldsymbol{w}]_S$ is the coordinate vector of $\boldsymbol{w}$ relative to a basis S. The transition matrix from T to S is the matrix $\boldsymbol{P}$ with $[\boldsymbol{w}]_S = \boldsymbol{P}[\boldsymbol{w}]_T$. Stochastic matrices have columns that sum to 1, so a Markov chain runs as $\boldsymbol{x}_{n+1} = \boldsymbol{P}\boldsymbol{x}_n$.

Results You Are Expected to Know Cold

INVERTIBILITY: EQUIVALENT STATEMENTS FOR A SQUARE MATRIX OF ORDER n

$\boldsymbol{A}$ is invertible $\Leftrightarrow$ the RREF of $\boldsymbol{A}$ is $\boldsymbol{I}$ $\Leftrightarrow$ $\boldsymbol{A}$ is a product of elementary matrices $\Leftrightarrow$ $\boldsymbol{Ax} = \boldsymbol{0}$ has only the trivial solution $\Leftrightarrow$ $\boldsymbol{Ax} = \boldsymbol{b}$ has a unique solution for every $\boldsymbol{b}$ $\Leftrightarrow$ $\det(\boldsymbol{A}) \neq 0$ $\Leftrightarrow$ the columns (rows) of $\boldsymbol{A}$ form a basis for $\mathbb{R}^n$ $\Leftrightarrow$ $\operatorname{rank}(\boldsymbol{A}) = n$ $\Leftrightarrow$ $\operatorname{nullity}(\boldsymbol{A}) = 0$ $\Leftrightarrow$ 0 is not an eigenvalue of $\boldsymbol{A}$.

DETERMINANTS

$\det(\boldsymbol{AB}) = \det(\boldsymbol{A})\det(\boldsymbol{B})$, $\det\left(\boldsymbol{A}^{\mathrm{T}}\right) = \det(\boldsymbol{A})$, $\det\left(\boldsymbol{A}^{-1}\right) = 1/\det(\boldsymbol{A})$, $\det(c\boldsymbol{A}) = c^n \det(\boldsymbol{A})$.
Row operations: $R_i \leftrightarrow R_j$ changes the sign; cR_i multiplies the determinant by c; $R_i + cR_j$ leaves it unchanged. A triangular matrix has determinant equal to the product of its diagonal entries. $\boldsymbol{A}^{-1} = \frac{1}{\det(\boldsymbol{A})}\operatorname{adj}(\boldsymbol{A})$.

RANK

For an $m \times n$ matrix, $\operatorname{rank}(\boldsymbol{A}) = \dim \operatorname{Col}(\boldsymbol{A}) = \dim \operatorname{Row}(\boldsymbol{A})$ is the number of pivot columns, and $\operatorname{rank}(\boldsymbol{A}) + \operatorname{nullity}(\boldsymbol{A}) = n$. Row operations preserve the row space and the nullspace, but not the column space. $\boldsymbol{Ax} = \boldsymbol{b}$ is consistent $\Leftrightarrow$ $\boldsymbol{b} \in \operatorname{Col}(\boldsymbol{A})$ $\Leftrightarrow$ $\operatorname{rank}(\boldsymbol{A}) = \operatorname{rank}(\boldsymbol{A} \mid \boldsymbol{b})$. $\operatorname{rank}(\boldsymbol{A}) = n$ $\Leftrightarrow$ $\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}$ is invertible $\Leftrightarrow$ $\boldsymbol{A}$ has a left inverse; $\operatorname{rank}(\boldsymbol{A}) = m$ $\Leftrightarrow$ $\boldsymbol{A}\boldsymbol{A}^{\mathrm{T}}$ is invertible $\Leftrightarrow$ $\boldsymbol{A}$ has a right inverse.

ORTHOGONALITY AND LEAST SQUARES

If $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k\}$ is an orthogonal basis for V, the projection of $\boldsymbol{w}$ onto V is

$$\boldsymbol{w}_p = \frac{\boldsymbol{w} \cdot \boldsymbol{u}_1}{\boldsymbol{u}_1 \cdot \boldsymbol{u}_1}\boldsymbol{u}_1 + \cdots + \frac{\boldsymbol{w} \cdot \boldsymbol{u}_k}{\boldsymbol{u}_k \cdot \boldsymbol{u}_k}\boldsymbol{u}_k.$$

Gram–Schmidt: $\boldsymbol{v}_1 = \boldsymbol{u}_1$, $\boldsymbol{v}_{i+1} = \boldsymbol{u}_{i+1} - \sum_{j \le i} \frac{\boldsymbol{u}_{i+1} \cdot \boldsymbol{v}_j}{\boldsymbol{v}_j \cdot \boldsymbol{v}_j}\boldsymbol{v}_j$. QR: $\boldsymbol{A} = \boldsymbol{QR}$ with $\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{Q} = \boldsymbol{I}$ and $\boldsymbol{R} = \boldsymbol{Q}^{\mathrm{T}}\boldsymbol{A}$ upper triangular.
$\hat{\boldsymbol{x}}$ is a least squares solution of $\boldsymbol{Ax} = \boldsymbol{b}$ $\Leftrightarrow$ $\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}\hat{\boldsymbol{x}} = \boldsymbol{A}^{\mathrm{T}}\boldsymbol{b}$ $\Leftrightarrow$ $\boldsymbol{A}\hat{\boldsymbol{x}}$ is the projection of $\boldsymbol{b}$ onto $\operatorname{Col}(\boldsymbol{A})$. With $\boldsymbol{A} = \boldsymbol{QR}$: $\boldsymbol{R}\hat{\boldsymbol{x}} = \boldsymbol{Q}^{\mathrm{T}}\boldsymbol{b}$.

DIAGONALIZATION

$\boldsymbol{A}$ of order n is diagonalizable $\Leftrightarrow$ it has n linearly independent eigenvectors $\Leftrightarrow$ the characteristic polynomial splits into linear factors and, for every eigenvalue, the geometric multiplicity $\dim \operatorname{Null}(\lambda\boldsymbol{I} - \boldsymbol{A})$ equals the algebraic multiplicity. Then $\boldsymbol{A} = \boldsymbol{PDP}^{-1}$ and $\boldsymbol{A}^k = \boldsymbol{PD}^k\boldsymbol{P}^{-1}$.
$\boldsymbol{A}$ is orthogonally diagonalizable ($\boldsymbol{A} = \boldsymbol{PDP}^{\mathrm{T}}$ with $\boldsymbol{P}$ orthogonal) $\Leftrightarrow$ $\boldsymbol{A}$ is symmetric.

SINGULAR VALUE DECOMPOSITION OF AN $m \times n$ MATRIX

$\boldsymbol{A} = \boldsymbol{U}\Sigma\boldsymbol{V}^{\mathrm{T}}$ with $\boldsymbol{U}$ $(m \times m)$ and $\boldsymbol{V}$ $(n \times n)$ orthogonal. The singular values $\sigma_1 \ge \cdots \ge \sigma_r > 0$ are the square roots of the nonzero eigenvalues of $\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}$; the columns $\boldsymbol{v}_i$ of $\boldsymbol{V}$ are orthonormal eigenvectors of $\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}$ in the same order; $\boldsymbol{u}_i = \frac{1}{\sigma_i}\boldsymbol{A}\boldsymbol{v}_i$ for $i \le r$, and $\boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_m$ extend these to an orthonormal basis of $\mathbb{R}^m$. Then $r = \operatorname{rank}(\boldsymbol{A})$, $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r\}$ is a basis for $\operatorname{Col}(\boldsymbol{A})$ and $\{\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_n\}$ is a basis for $\operatorname{Null}(\boldsymbol{A})$.

PART A

Problems 1–10

Linear Systems, Matrices, and Determinants

Chapters 1 and 2: row reduction, inverses, elementary matrices, LU factorization, determinants

Solving Linear Systems

1. Use Gauss–Jordan elimination to find the general solution of

$$\begin{aligned} x_1 + 2x_2 - x_3 + x_4 &= 3 \\ 2x_1 + 4x_2 - x_3 + 4x_4 &= 8 \\ -x_1 - 2x_2 + 3x_3 + 3x_4 &= 1. \end{aligned}$$

State which variables are assigned parameters and why.

2. Let a be a real constant. Determine the values of a for which the system

$$x + y + az = 1, \quad x + ay + z = 1, \quad ax + y + z = -2$$

has (i) no solution, (ii) exactly one solution, (iii) infinitely many solutions. In case (iii), write down the general solution.

3. Let c and d be real constants. Find the rank and the nullity of

$$\begin{pmatrix} 1 & 2 & -1 & 3 \\ 2 & c & -2 & c+2 \\ -1 & -2 & d & d-4 \end{pmatrix}$$

for all possible values of c and d.

Inverses, Elementary Matrices, and LU Factorization

4. Let $\boldsymbol{A} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 1 & 3 & 4 \end{pmatrix}$ and $\boldsymbol{B} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 3 \end{pmatrix}$.

(a) Find $\boldsymbol{A}^{-1}$ by row reducing $(\boldsymbol{A} \mid \boldsymbol{I})$.

(b) Hence solve the matrix equation $\boldsymbol{AX} = \boldsymbol{B}$.

5. Let $\boldsymbol{A}$ be a square matrix of order 3 with $\det(\boldsymbol{A}) = 4$. The matrix $\boldsymbol{B}$ is obtained from $\boldsymbol{A}$ by performing, in order, the row operations $R_1 \leftrightarrow R_3$, $3R_2$, $R_3 + 2R_1$.

(a) Write $\boldsymbol{B} = \boldsymbol{E}_3\boldsymbol{E}_2\boldsymbol{E}_1\boldsymbol{A}$ and write down the elementary matrices $\boldsymbol{E}_1, \boldsymbol{E}_2, \boldsymbol{E}_3$.

(b) Find $\det(\boldsymbol{B})$ and $\det\left(2\boldsymbol{A}^{-1}\boldsymbol{B}^{\mathrm{T}}\right)$.

6. Let $\boldsymbol{A} = \begin{pmatrix} 2 & 1 & -1 \\ 4 & 5 & -3 \\ -2 & 5 & 3 \end{pmatrix}$.

(a) Find an LU factorization $\boldsymbol{A} = \boldsymbol{L}\boldsymbol{U}$, with $\boldsymbol{L}$ unit lower triangular.

(b) Use it to solve $\boldsymbol{A}\boldsymbol{x} = (1, 7, 13)^{\mathrm{T}}$ by solving $\boldsymbol{L}\boldsymbol{y} = \boldsymbol{b}$ and then $\boldsymbol{U}\boldsymbol{x} = \boldsymbol{y}$.

Determinants

7. For $n \geq 1$ let $\boldsymbol{D}_n$ be the square matrix of order n with every diagonal entry equal to 2, every entry immediately above or below the diagonal equal to 1, and all other entries 0, and let $d_n = \det(\boldsymbol{D}_n)$.

(a) Compute d_1, d_2 and d_3.

(b) By cofactor expansion along the first row, show that $d_n = 2d_{n-1} - d_{n-2}$ for $n \geq 3$.

(c) Deduce a formula for d_n. For which n is $\boldsymbol{D}_n$ invertible?

8. Let t be a real constant and $\boldsymbol{M} = \begin{pmatrix} 1 & t & 0 \\ t & 1 & t \\ 0 & t & 1 \end{pmatrix}$. Find $\det(\boldsymbol{M})$, and all values of t for which the homogeneous system $\boldsymbol{M}\boldsymbol{x} = \boldsymbol{0}$ has a nontrivial solution.

9. Let $\boldsymbol{A}$ be a square matrix of order 3 with $\det(\boldsymbol{A}) = 2$. Find $\det\left(\boldsymbol{A}^3\right)$, $\det\left(3\boldsymbol{A}^{-1}\right)$, $\det(\operatorname{adj}(\boldsymbol{A}))$ and $\det\left(\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A} - 2\boldsymbol{A}^{\mathrm{T}}\right)$ given further that $\det(\boldsymbol{A} - 2\boldsymbol{I}) = 5$.

True or False

10. For each statement about square matrices $\boldsymbol{A}$, $\boldsymbol{B}$, $\boldsymbol{C}$ of order n, decide whether it is always true. Justify a true statement in one line and give a 2×2 counterexample for a false one.

(a) If $\boldsymbol{AB} = \mathbf{0}$, then $\boldsymbol{A} = \mathbf{0}$ or $\boldsymbol{B} = \mathbf{0}$.

(b) $(\boldsymbol{A} + \boldsymbol{B})^2 = \boldsymbol{A}^2 + 2\boldsymbol{AB} + \boldsymbol{B}^2$.

(c) If $\boldsymbol{A}^2 = \boldsymbol{I}$, then $\boldsymbol{A} = \boldsymbol{I}$ or $\boldsymbol{A} = -\boldsymbol{I}$.

(d) If $\boldsymbol{A}$ and $\boldsymbol{B}$ are symmetric, then $\boldsymbol{AB}$ is symmetric.

(e) If $\boldsymbol{A}$ is invertible and $\boldsymbol{AB} = \boldsymbol{AC}$, then $\boldsymbol{B} = \boldsymbol{C}$.

PART B

Problems 11–18

Euclidean Vector Spaces

Chapter 3: spans, subspaces, linear independence, bases, coordinates, transition matrices

Spans and Subspaces

11. Let $S = \{(1,0,1,2)^{\mathrm{T}}, (0,1,1,-1)^{\mathrm{T}}, (1,1,0,1)^{\mathrm{T}}\}$, $\boldsymbol{v} = (3,1,2,5)^{\mathrm{T}}$ and $\boldsymbol{w} = (1,1,1,0)^{\mathrm{T}}$.

(a) Decide whether each of $\boldsymbol{v}$ and $\boldsymbol{w}$ belongs to $\operatorname{span}(S)$.

(b) For each vector that does, write down its coordinates relative to S.

12. Decide whether each set is a subspace. Express a subspace as a span, and for a set that is not a subspace name a condition that fails, with specific vectors.

(a) $\{(x,y,z)^{\mathrm{T}} \in \mathbb{R}^3 : x - 2y + z = 0\}$

(b) $\{(x,y,z)^{\mathrm{T}} \in \mathbb{R}^3 : x + y + z = 1\}$

(c) $\{(a+b, a-b, 2a)^{\mathrm{T}} : a, b \in \mathbb{R}\}$

(d) $\{(x,y)^{\mathrm{T}} \in \mathbb{R}^2 : xy \geq 0\}$

(e) $\{(x,y,z)^{\mathrm{T}} \in \mathbb{R}^3 : x^2 + y^2 = 0\}$

Linear Independence and Bases

13. Find all real k for which $\{(1,k,0)^{\mathrm{T}},(0,1,k)^{\mathrm{T}},(k,0,1)^{\mathrm{T}}\}$ is linearly dependent, and for each such k write one of the vectors as a linear combination of the other two.

14. Let $V=\{\boldsymbol{x}\in\mathbb{R}^4 : x_1-x_2+2x_3=0 \text{ and } x_2+x_3-x_4=0\}$.

(a) Find a basis for V and state $\dim V$.

(b) Extend your basis to a basis for $\mathbb{R}^4$, and justify that it is one.

15. Let $\boldsymbol{u}_1=(1,2,0,1)^{\mathrm{T}}$, $\boldsymbol{u}_2=(2,4,0,2)^{\mathrm{T}}$, $\boldsymbol{u}_3=(0,1,1,1)^{\mathrm{T}}$, $\boldsymbol{u}_4=(1,3,1,2)^{\mathrm{T}}$, $\boldsymbol{u}_5=(1,0,1,0)^{\mathrm{T}}$, and $W=\operatorname{span}\{\boldsymbol{u}_1,\ldots,\boldsymbol{u}_5\}$.

(a) Find a subset of $\{\boldsymbol{u}_1,\ldots,\boldsymbol{u}_5\}$ that is a basis for W.

(b) Write each of the remaining vectors as a linear combination of your basis.

Coordinates and Transition Matrices

16. Let $S = \{(1,0,1)^{\mathrm{T}}, (1,1,0)^{\mathrm{T}}, (0,1,1)^{\mathrm{T}}\}$ and $T = \{(1,1,1)^{\mathrm{T}}, (1,1,0)^{\mathrm{T}}, (1,0,0)^{\mathrm{T}}\}$, two bases for $\mathbb{R}^3$.

(a) Find the transition matrix $\boldsymbol{P}$ from T to S.

(b) A vector $\boldsymbol{w}$ has $[\boldsymbol{w}]_T = (1,2,-1)^{\mathrm{T}}$. Find $[\boldsymbol{w}]_S$ and $\boldsymbol{w}$.

(c) Write down the transition matrix from S to T.

Theory

17. Let U and V be subspaces of $\mathbb{R}^5$ with $\dim U = 3$ and $\dim V = 4$. Using $\dim(U+V) = \dim U + \dim V - \dim(U \cap V)$, find all possible values of $\dim(U \cap V)$, and give the value when $U \not\subseteq V$.

18. Let $\{\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}\}$ be a linearly independent set in $\mathbb{R}^n$.

(a) Prove that $\{\boldsymbol{u}+\boldsymbol{v}, \boldsymbol{v}+\boldsymbol{w}, \boldsymbol{w}+\boldsymbol{u}\}$ is linearly independent.

(b) Show that $\{\boldsymbol{u}-\boldsymbol{v}, \boldsymbol{v}-\boldsymbol{w}, \boldsymbol{w}-\boldsymbol{u}\}$ is linearly dependent.

PART C

Problems 19–24

Subspaces Associated to a Matrix and Rank

Chapter 4: column space, row space, nullspace, rank–nullity, full rank

Bases for the Fundamental Subspaces

19. Let $A = \begin{pmatrix} 1 & 2 & 0 & 1 & -1 \\ 2 & 4 & 1 & 4 & -1 \\ -1 & -2 & 1 & 1 & 2 \\ 3 & 6 & 1 & 5 & -2 \end{pmatrix}$.

(a) Find the reduced row-echelon form of A.

(b) Find a basis for each of $\mathrm{Row}(A)$, $\mathrm{Col}(A)$ and $\mathrm{Null}(A)$.

(c) State $\mathrm{rank}(A)$, $\mathrm{nullity}(A)$ and $\mathrm{nullity}\left(A^{\mathrm{T}}\right)$.

20. Let A be a 3×5 matrix. The system $Ax = b$ has general solution

$$x = (1 + s - 2t,\ s,\ 2 + t,\ t,\ 4)^{\mathrm{T}}, \quad s, t \in \mathbb{R}.$$

(a) Find $\mathrm{nullity}(A)$, $\mathrm{rank}(A)$ and a basis for $\mathrm{Null}(A)$.

(b) Is $Ax = c$ consistent for every $c \in \mathbb{R}^3$? Does A have a left inverse? A right inverse?

21. Let $\boldsymbol{A} = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \end{pmatrix}$.

(a) Explain why $\boldsymbol{A}$ has a right inverse but no left inverse.

(b) Find all right inverses of $\boldsymbol{A}$.

22. Let $\boldsymbol{A} = \begin{pmatrix} 1 & -1 & 2 \\ 2 & 1 & 1 \\ 4 & -1 & 5 \end{pmatrix}$. Find a condition on $\boldsymbol{b} = (b_1, b_2, b_3)^{\mathrm{T}}$ that is necessary and sufficient for $\boldsymbol{Ax} = \boldsymbol{b}$ to be consistent, and hence describe $\mathrm{Col}(\boldsymbol{A})$ as the solution set of a homogeneous linear equation.

Theory

23. Decide whether each statement is always true, with a brief justification or a counterexample.

(a) $\operatorname{rank}(\boldsymbol{A}) = \operatorname{rank}\left(\boldsymbol{A}^{\mathrm{T}}\right)$.

(b) If $\boldsymbol{A}$ and $\boldsymbol{B}$ are row equivalent, then $\operatorname{Col}(\boldsymbol{A}) = \operatorname{Col}(\boldsymbol{B})$.

(c) $\operatorname{rank}(\boldsymbol{AB}) \leq \operatorname{rank}(\boldsymbol{A})$ whenever the product is defined.

(d) There is a 5×7 matrix with nullity 1.

(e) If $\boldsymbol{Ax} = \boldsymbol{b}$ is consistent for every $\boldsymbol{b}$, then $\operatorname{rank}(\boldsymbol{A})$ equals the number of rows of $\boldsymbol{A}$.

24. Let $\boldsymbol{A}$ be an $m \times n$ matrix.

(a) Prove that $\operatorname{Null}\left(\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}\right) = \operatorname{Null}(\boldsymbol{A})$. (Hint: consider $\boldsymbol{x}^{\mathrm{T}}\boldsymbol{A}^{\mathrm{T}}\boldsymbol{Ax}$.)

(b) Deduce that $\operatorname{rank}\left(\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}\right) = \operatorname{rank}(\boldsymbol{A})$, and that $\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}$ is invertible exactly when the columns of $\boldsymbol{A}$ are linearly independent.

PART D

Problems 25–33

Orthogonality and Least Squares

Chapter 5: orthogonal bases, projections, Gram–Schmidt, QR factorization, least squares

Orthogonal Sets and Projections

25. Let $\boldsymbol{u}_1 = (1,1,1,1)^{\mathrm{T}}$, $\boldsymbol{u}_2 = (1,-1,1,-1)^{\mathrm{T}}$, $\boldsymbol{u}_3 = (1,1,-1,-1)^{\mathrm{T}}$ and $V = \operatorname{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\}$.

(a) Verify that $\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\}$ is an orthogonal set, and explain why it is a basis for V.

(b) Show that $\boldsymbol{w} = (4,2,0,-2)^{\mathrm{T}}$ lies in V by finding its coordinates relative to this basis using dot products only.

(c) Find the projection of $\boldsymbol{v} = (1,0,0,0)^{\mathrm{T}}$ onto V and the distance from $\boldsymbol{v}$ to V.

26. Let $V = \operatorname{span}\left\{(1,1,0,-1)^{\mathrm{T}}, (0,1,2,1)^{\mathrm{T}}\right\}$. Find a basis for the orthogonal complement $V^{\perp}$, and verify that $\dim V + \dim V^{\perp} = 4$.

27. Let $V = \operatorname{span}\{(1,2,2)^{\mathrm{T}}\}$ in $\mathbb{R}^3$.

(a) Find the matrix $\boldsymbol{P}$ such that $\boldsymbol{P}\boldsymbol{x}$ is the projection of $\boldsymbol{x}$ onto V for every $\boldsymbol{x} \in \mathbb{R}^3$.

(b) Find the matrix of the projection onto $V^{\perp}$.

(c) Without computing, state the eigenvalues of $\boldsymbol{P}$ and their eigenspaces.

Gram–Schmidt Process and QR Factorization

28. Apply the Gram–Schmidt process to

$$\boldsymbol{u}_1 = (1,1,0,0)^{\mathrm{T}}, \quad \boldsymbol{u}_2 = (1,0,1,0)^{\mathrm{T}}, \quad \boldsymbol{u}_3 = (0,1,1,1)^{\mathrm{T}}$$

to obtain an orthonormal basis for $\operatorname{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\}$.

29. Let $A = \begin{pmatrix} 1 & -1 \\ 2 & 4 \\ 2 & 1 \end{pmatrix}$ and $b = (3, 0, 3)^{\mathrm{T}}$.

(a) Find a QR factorization $A = QR$, where Q has orthonormal columns and R is upper triangular with positive diagonal entries.

(b) Use it to find the least squares solution of $Ax = b$.

30. Let $Q = \frac{1}{3}\begin{pmatrix} 2 & -2 & 1 \\ 1 & 2 & 2 \\ 2 & 1 & -2 \end{pmatrix}$.

(a) Show that Q is an orthogonal matrix, and find $\det(Q)$.

(b) Solve $Qx = (3, 6, 9)^{\mathrm{T}}$ without row reduction.

Least Squares

31. Find the line $y = a + bx$ that best fits the data points $(-1, 0)$, $(0, 1)$, $(1, 3)$, $(2, 4)$ in the least squares sense. Set up $\boldsymbol{Ax} = \boldsymbol{b}$ and solve the normal equations.

32. Let $\boldsymbol{A} = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$ and $\boldsymbol{b} = (1, 2, 3)^{\mathrm{T}}$.

(a) Show that $\boldsymbol{Ax} = \boldsymbol{b}$ is inconsistent.

(b) Find all least squares solutions of $\boldsymbol{Ax} = \boldsymbol{b}$.

(c) Find the projection of $\boldsymbol{b}$ onto $\mathrm{Col}(\boldsymbol{A})$, and explain why it does not depend on the least squares solution chosen.

Theory

33. Let Q be an $m \times n$ matrix with orthonormal columns.

(a) Prove that $\|Q\boldsymbol{x}\| = \|\boldsymbol{x}\|$ and $(Q\boldsymbol{x}) \cdot (Q\boldsymbol{y}) = \boldsymbol{x} \cdot \boldsymbol{y}$ for all $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$.

(b) Show that $QQ^{\mathrm{T}}\boldsymbol{b}$ is the projection of $\boldsymbol{b}$ onto $\mathrm{Col}(Q)$ for every $\boldsymbol{b} \in \mathbb{R}^m$.

PART E

Problems 34–46

Eigenvalues, Diagonalization, and the SVD

Chapter 6: eigenspaces, diagonalization, orthogonal diagonalization, Markov chains, singular value decomposition

Eigenvalues and Diagonalization

34. Let $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ -1 & 1 & 2 \end{pmatrix}$.

(a) Find the characteristic polynomial and the eigenvalues of A.

(b) Find a basis for each eigenspace, and state the algebraic and geometric multiplicity of each eigenvalue.

(c) Find an invertible P and a diagonal D with $A = PDP^{-1}$.

35. Let a, b, c be real constants and $C = \begin{pmatrix} 1 & a & b \\ 0 & 2 & c \\ 0 & 0 & 1 \end{pmatrix}$. Find a condition on a, b, c that is necessary and sufficient for C to be diagonalizable.

36. Let $A = \begin{pmatrix} 4 & -2 \\ 1 & 1 \end{pmatrix}$.

(a) Diagonalize A.

(b) Find a formula for A^n, for every integer $n \geq 1$, and check it at $n = 1$.

37. A square matrix A of order 3 has eigenvalues 1, -2 and 3. Find $\det(A)$, the trace of A, the eigenvalues of A^{-1} and of $A^2 + I$, and $\det(A^2 - A)$. Is $A + 2I$ invertible?

Orthogonal Diagonalization

38. Let $\boldsymbol{A} = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{pmatrix}$. Find an orthogonal matrix $\boldsymbol{P}$ and a diagonal matrix $\boldsymbol{D}$ such that $\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{\mathrm{T}}$.

39. A symmetric matrix $\boldsymbol{A}$ of order 3 has eigenvalues 1 and 4. The eigenspace for the eigenvalue 1 is $\operatorname{span}\{(1,1,1)^{\mathrm{T}}\}$.

(a) Describe the eigenspace for the eigenvalue 4, with justification.

(b) Find $\boldsymbol{A}$.

Markov Chains

40. A bike-share scheme has three docking stations X, Y, Z. Of the bikes at X in the morning, 50% are back at X the next morning, 30% are at Y and 20% at Z. Of those at Y, 20% move to X, 60% stay and 20% move to Z. Of those at Z, 30% move to X, 30% to Y and 40% stay.

(a) Write down the stochastic matrix $\boldsymbol{P}$ of this Markov chain.

(b) A bike is at X this morning. Find the probability that it is at Y two mornings later.

(c) Find the equilibrium vector, and interpret it.

41. Let $\boldsymbol{P} = \begin{pmatrix} 0.9 & 0.4 \\ 0.1 & 0.6 \end{pmatrix}$ and $\boldsymbol{x}_0 = (1, 0)^{\mathrm{T}}$, and let $\boldsymbol{x}_n = \boldsymbol{P}^n \boldsymbol{x}_0$.

(a) Diagonalize $\boldsymbol{P}$.

(b) Find $\boldsymbol{x}_n$ explicitly, and $\lim_{n\to\infty} \boldsymbol{x}_n$.

Singular Value Decomposition

42. Find a singular value decomposition $\boldsymbol{A} = \boldsymbol{U}\Sigma\boldsymbol{V}^{\mathrm{T}}$ of $\boldsymbol{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{pmatrix}$.

43. Find a singular value decomposition of $\boldsymbol{A} = \begin{pmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix}$. (It is quicker to start from $\boldsymbol{A}\boldsymbol{A}^{\mathrm{T}}$, or equivalently to decompose $\boldsymbol{A}^{\mathrm{T}}$ first.)

44. A 4×3 matrix has the singular value decomposition $\boldsymbol{A} = \boldsymbol{U}\Sigma\boldsymbol{V}^{\mathrm{T}}$ with

$$\boldsymbol{U} = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 12 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \boldsymbol{V} = \frac{1}{3}\begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix}.$$

(a) Find $\operatorname{rank}(\boldsymbol{A})$ and the eigenvalues of $\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}$.

(b) Find the general solution of $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}$.

(c) Find the projection of $\boldsymbol{w} = (1, 2, 3, 4)^{\mathrm{T}}$ onto $\operatorname{Col}(\boldsymbol{A})$.

(d) Find the matrix $\boldsymbol{A}$.

Theory

45. Decide whether each statement about a square matrix is always true, with a brief justification or a counterexample.

(a) Every invertible matrix is diagonalizable.

(b) Every diagonalizable matrix is invertible.

(c) A matrix of order n with n distinct eigenvalues is diagonalizable.

(d) If $\boldsymbol{A}$ is diagonalizable, then so is $\boldsymbol{A}^{\mathrm{T}}$.

(e) Two matrices with the same characteristic polynomial are either both diagonalizable or both not diagonalizable.

46. Let $\boldsymbol{A}$ and $\boldsymbol{B}$ be square matrices of order n.

(a) Prove that if $\boldsymbol{A}^2 = \boldsymbol{A}$, then every eigenvalue of $\boldsymbol{A}$ is 0 or 1.

(b) Prove that if $\lambda \neq 0$ is an eigenvalue of $\boldsymbol{AB}$, then λ is an eigenvalue of $\boldsymbol{BA}$.

PART F

Problems 47–54

Linear Transformations

Chapter 7: standard matrix, representation with respect to a basis, range and kernel

Linearity and the Standard Matrix

47. Decide whether each mapping is a linear transformation. Give the standard matrix if it is, and a specific failure of linearity if it is not. Here $\boldsymbol{a}$ is a fixed vector in $\mathbb{R}^3$.

(a) $T(x, y) = (x + 2y, 3y, x)$

(b) $T(x, y) = (x + 1, y)$

(c) $T(x, y) = (xy, x)$

(d) $T(\boldsymbol{x}) = (\boldsymbol{x} \cdot \boldsymbol{a})\boldsymbol{a}$ for $\boldsymbol{x} \in \mathbb{R}^3$, with $\boldsymbol{a} = (1, 0, 2)^{\mathrm{T}}$

48. A linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^2$ satisfies

$$T\begin{pmatrix}1\\1\\0\end{pmatrix} = \begin{pmatrix}1\\2\end{pmatrix}, \quad T\begin{pmatrix}0\\1\\1\end{pmatrix} = \begin{pmatrix}0\\1\end{pmatrix}, \quad T\begin{pmatrix}1\\0\\1\end{pmatrix} = \begin{pmatrix}3\\-1\end{pmatrix}.$$

(a) Write down the representation $[T]_S$ of T with respect to the basis $S = \{(1, 1, 0)^{\mathrm{T}}, (0, 1, 1)^{\mathrm{T}}, (1, 0, 1)^{\mathrm{T}}\}$.

(b) Find the standard matrix of T, and hence $T(x, y, z)$.

49. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the reflection about the line $y = 2x$.

(a) By considering $T(\boldsymbol{u})$ for $\boldsymbol{u}$ on the line and for $\boldsymbol{u}$ perpendicular to it, write down $[T]_S$ for $S = \left\{(1,2)^{\mathrm{T}}, (2,-1)^{\mathrm{T}}\right\}$.

(b) Find the standard matrix $\boldsymbol{A}$ of T. What are $\boldsymbol{A}^2$ and $\det(\boldsymbol{A})$?

50. Let $R : \mathbb{R}^2 \to \mathbb{R}^2$ be the anticlockwise rotation about the origin by 90° and $Q : \mathbb{R}^2 \to \mathbb{R}^2$ the projection onto the x-axis. Find the standard matrices of $Q \circ R$ and of $R \circ Q$, and the kernel and range of $Q \circ R$.

Range and Kernel

51. Let $T : \mathbb{R}^4 \to \mathbb{R}^3$ be given by

$$T(\boldsymbol{x}) = (x_1 + 2x_2 - x_4,\ x_2 + x_3 + x_4,\ x_1 + 3x_2 + x_3)^{\mathrm{T}}.$$

(a) Find a basis for the kernel of T and a basis for the range of T.

(b) State rank(T) and nullity(T). Is T injective? Is T surjective?

52. Let a be a real constant and $T(x, y, z) = (ax + y,\ x + ay + z,\ y + az)$.

(a) Find all values of a for which the kernel of T is nontrivial.

(b) For the rational such value of a, find a basis for $\ker(T)$.

53. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ have standard matrix $\boldsymbol{A} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & \frac{1}{2} & 1 \\ 0 & 0 & \frac{1}{3} \end{pmatrix}$, and write T^n for T composed with itself n times.

(a) Find the eigenvalues of $\boldsymbol{A}$ and an eigenvector for each.

(b) For $\boldsymbol{x} = (x_1, x_2, x_3)^{\mathrm{T}}$, find $\lim_{n\to\infty} T^n(\boldsymbol{x})$ in terms of x_1, x_2, x_3.

Theory

54. Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation.

(a) Prove that T is injective if and only if $\ker(T) = \{\mathbf{0}\}$.

(b) Prove that if $n > m$ then T is not injective, and that if $n < m$ then T is not surjective.

PART G

Problems 55–75 • 100 marks

Mock Final Examination

16 multiple-choice questions of 2 marks each and 5 written questions worth 68 marks • 2 hours, closed book, one A4 helpsheet

INSTRUCTIONS

Sit this paper in one sitting of two hours, with one double-sided A4 helpsheet and an approved calculator, and nothing else.

Section A (Problems 55–70) has 16 multiple-choice questions of 2 marks each. Exactly one option is correct. In the examination the answers are shaded on an OMR sheet with a dark pencil; here, circle the letter.

Section B (Problems 71–75) has 5 written questions worth 68 marks in total. The marks for each part are shown in brackets. Write each answer in the space provided and show full working for all computations: an answer without working does not earn full credit.

Section A Multiple Choice (32 marks)

55. Which of the following matrices are in reduced row-echelon form?

$$\text{(I)}\begin{pmatrix}1&2&0\\0&0&1\end{pmatrix}\quad \text{(II)}\begin{pmatrix}1&0&3\\0&2&1\end{pmatrix}\quad \text{(III)}\begin{pmatrix}0&1&0\\0&0&0\\0&0&1\end{pmatrix}\quad \text{(IV)}\begin{pmatrix}1&0&0\\0&1&5\end{pmatrix}$$

(A) Only (I) and (IV).
(B) Only (I), (II) and (IV).
(C) Only (I), (III) and (IV).
(D) None of the above.

56. Consider a homogeneous linear system with 4 equations and 6 variables. Which of the following statements is/are always true?

(I) The system has a nontrivial solution.
(II) The general solution has exactly 2 arbitrary parameters.

(A) Both (I) and (II) are true.
(B) Only (I) is true.
(C) Only (II) is true.
(D) Both (I) and (II) are false.

57. Let $\boldsymbol{A}$ be a square matrix of order 3 with $\det(\boldsymbol{A}) = 5$. The matrix $\boldsymbol{B}$ is obtained from $\boldsymbol{A}$ by swapping two rows and then multiplying one row by 2. What is $\det\left(\boldsymbol{A}^{-1}\boldsymbol{B}^{\mathrm{T}}\right)$?

(A) 2
(B) $-\frac{1}{2}$
(C) -2
(D) -50

58. Let $\boldsymbol{A}$ and $\boldsymbol{B}$ be square matrices of order n. Which of the following statements is/are always true?

(I) If $\boldsymbol{A}+\boldsymbol{B}$ is invertible, then $\boldsymbol{A}$ is invertible or $\boldsymbol{B}$ is invertible.
(II) If $\boldsymbol{AB}$ is invertible, then $\boldsymbol{A}$ and $\boldsymbol{B}$ are both invertible.

(A) Both (I) and (II) are true.
(B) Only (I) is true.
(C) Only (II) is true.
(D) Both (I) and (II) are false.

59. Which of the following subsets of $\mathbb{R}^3$ is/are subspaces?

(I) $\left\{(x,y,z)^{\mathrm{T}} : x+y+z=1\right\}$
(II) $\left\{(x,y,z)^{\mathrm{T}} : x=2y\right\}$
(III) $\left\{(x,y,z)^{\mathrm{T}} : xy=0\right\}$

(A) Only (I).
(B) Only (II) and (III).
(C) Only (III).
(D) None of the above.

60. Let $\{\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}\}$ be a linearly independent subset of $\mathbb{R}^n$. Which of the following statements is/are always true?

(I) $\{\boldsymbol{u} - \boldsymbol{v}, \boldsymbol{v} - \boldsymbol{w}, \boldsymbol{w} - \boldsymbol{u}\}$ is linearly independent.
(II) $\{\boldsymbol{u} + \boldsymbol{v}, \boldsymbol{v} + \boldsymbol{w}, \boldsymbol{w} + \boldsymbol{u}\}$ is linearly independent.

(A) Both (I) and (II) are true. **(B)** Only (I) is true.
(C) Only (II) is true. **(D)** Both (I) and (II) are false.

61. Let $\boldsymbol{A}$ be a 4×7 matrix of rank 3. Which of the following is true?

(A) $\text{nullity}(\boldsymbol{A}) = 3$ and $\text{nullity}\left(\boldsymbol{A}^{\mathrm{T}}\right) = 1$. **(B)** $\text{nullity}(\boldsymbol{A}) = 4$ and $\text{nullity}\left(\boldsymbol{A}^{\mathrm{T}}\right) = 0$.
(C) $\text{nullity}(\boldsymbol{A}) = 3$ and $\text{nullity}\left(\boldsymbol{A}^{\mathrm{T}}\right) = 4$. **(D)** $\text{nullity}(\boldsymbol{A}) = 4$ and $\text{nullity}\left(\boldsymbol{A}^{\mathrm{T}}\right) = 1$.

62. Let $\boldsymbol{A}$ and $\boldsymbol{B}$ be row equivalent matrices. Which of the following statements is/are always true?

(I) $\text{Row}(\boldsymbol{A}) = \text{Row}(\boldsymbol{B})$.
(II) $\text{Col}(\boldsymbol{A}) = \text{Col}(\boldsymbol{B})$.
(III) $\text{Null}(\boldsymbol{A}) = \text{Null}(\boldsymbol{B})$.

(A) Only (I) is true. **(B)** Only (I) and (II) are true.
(C) Only (I) and (III) are true. **(D)** (I), (II) and (III) are all true.

63. Let S be an orthogonal set of k nonzero vectors in $\mathbb{R}^n$. Which of the following statements is/are always true?

(I) S is linearly independent.
(II) $k \leq n$.

(A) Both (I) and (II) are true. **(B)** Only (I) is true.
(C) Only (II) is true. **(D)** Both (I) and (II) are false.

64. Let $\boldsymbol{A}$ be an $m \times n$ matrix and $\boldsymbol{b} \in \mathbb{R}^m$. Which of the following statements is/are always true?

(I) $\boldsymbol{Ax} = \boldsymbol{b}$ has a least squares solution.
(II) The least squares solution of $\boldsymbol{Ax} = \boldsymbol{b}$ is unique.
(III) $\boldsymbol{A}\hat{\boldsymbol{x}}$ is the same vector for every least squares solution $\hat{\boldsymbol{x}}$.

(A) Only (I) and (III) are true. **(B)** Only (I) and (II) are true.
(C) Only (I) is true. **(D)** (I), (II) and (III) are all true.

65. What is the projection of $(1, 2, 3)^{\mathrm{T}}$ onto $\text{span}\left\{(1, 1, 1)^{\mathrm{T}}\right\}$?

(A) $(1, 1, 1)^{\mathrm{T}}$ **(B)** $(2, 2, 2)^{\mathrm{T}}$
(C) $(3, 3, 3)^{\mathrm{T}}$ **(D)** $(6, 6, 6)^{\mathrm{T}}$

66. Which of the following matrices is/are diagonalizable?

$$\text{(I)} \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} \quad \text{(II)} \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} \quad \text{(III)} \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

(A) Only (I). **(B)** Only (III).
(C) Only (II) and (III). **(D)** Only (I) and (III).

67. Let $\boldsymbol{A}$ be a square matrix of order 3 whose characteristic polynomial is $x^2(x - 3)$. Which of the following statements is/are always true?

(I) $\boldsymbol{A}$ is singular.
(II) $\boldsymbol{A}$ is diagonalizable.
(III) $\text{rank}(\boldsymbol{A}) = 1$.

(A) Only (I) is true. **(B)** Only (I) and (II) are true.
(C) Only (I) and (III) are true. **(D)** (I), (II) and (III) are all true.

68. What is the equilibrium vector of the stochastic matrix $\boldsymbol{P} = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix}$?

(A) $(0.5, 0.5)^{\mathrm{T}}$
(B) $(0.6, 0.4)^{\mathrm{T}}$
(C) $(0.3, 0.7)^{\mathrm{T}}$
(D) $(0.4, 0.6)^{\mathrm{T}}$

69. Let $\boldsymbol{A}$ be a 3×2 matrix with $\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$. What are the singular values of $\boldsymbol{A}$?

(A) 9 and 1
(B) 3 and 1
(C) $\sqrt{5}$ and 2
(D) 5 and 4

70. Let $T : \mathbb{R}^5 \to \mathbb{R}^3$ be a linear transformation. Which of the following statements is/are always true?

(I) T is not injective.
(II) T is surjective.

(A) Both (I) and (II) are true.
(B) Only (I) is true.
(C) Only (II) is true.
(D) Both (I) and (II) are false.

Section B Written Questions (68 marks)

71. [14 marks] Let a be a real constant and $\boldsymbol{A} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & a \\ 1 & a & 2 \end{pmatrix}$.

(a) Compute $\det(\boldsymbol{A})$ in terms of a. [3]

(b) For which values of a is $\boldsymbol{A}$ invertible? [1]

(c) For $a = 1$, find $\boldsymbol{A}^{-1}$. [4]

(d) For $a = 2$, find all values of c for which $\boldsymbol{A}\boldsymbol{x} = (1, 3, c)^{\mathrm{T}}$ is consistent, and find the general solution for those values. [6]

72. [14 marks] Let $V = \{\boldsymbol{x} \in \mathbb{R}^4 : x_1 - x_2 + x_3 - x_4 = 0\}$ and

$$S = \{\boldsymbol{u}_1 = (1, 1, 0, 0)^{\mathrm{T}},\ \boldsymbol{u}_2 = (0, 1, 1, 0)^{\mathrm{T}},\ \boldsymbol{u}_3 = (0, 0, 1, 1)^{\mathrm{T}}\}.$$

(a) Show that S is a basis for V. [4]

(b) Show that $\boldsymbol{w} = (2, 3, 4, 3)^{\mathrm{T}}$ lies in V, and find $[\boldsymbol{w}]_S$. [3]

(c) Find the projection of $\boldsymbol{y} = (3, -1, 1, 1)^{\mathrm{T}}$ onto V. [3]

(d) Let $T : \mathbb{R}^4 \to \mathbb{R}^4$ map each vector to its projection onto V. Find the standard matrix of T. [2]

(e) Write down $\ker(T)$, the range of T, and the rank and nullity of T. [2]

73. [12 marks] Let $\boldsymbol{A} = \begin{pmatrix} 1 & 3 \\ 1 & 1 \\ 1 & 3 \\ 1 & 1 \end{pmatrix}$ and $\boldsymbol{b} = (3, 1, 5, -1)^{\mathrm{T}}$.

(a) Find a matrix $\boldsymbol{Q}$ with orthonormal columns and an upper triangular matrix $\boldsymbol{R}$ with positive diagonal entries such that $\boldsymbol{A} = \boldsymbol{QR}$. [6]

(b) Find the least squares solution of $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$. [3]

(c) Find the projection of $\boldsymbol{b}$ onto $\mathrm{Col}(\boldsymbol{A})$ and the distance from $\boldsymbol{b}$ to $\mathrm{Col}(\boldsymbol{A})$. [3]

74. [12 marks] Let $\boldsymbol{A} = \begin{pmatrix} 1 & 2 \\ 1 & 0 \end{pmatrix}$, and define $\boldsymbol{x}_0 = (1, 0)^{\mathrm{T}}$ and $\boldsymbol{x}_{n+1} = \boldsymbol{A}\boldsymbol{x}_n$ for $n \geq 0$.

(a) Find the eigenvalues of $\boldsymbol{A}$ and a basis for each eigenspace. [4]

(b) Write down an invertible $\boldsymbol{P}$ and a diagonal $\boldsymbol{D}$ with $\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{-1}$. [2]

(c) Find A^n for every integer $n \geq 1$. [4]

(d) Find x_n, and the limit of the ratio of its first coordinate to its second as $n \to \infty$. [2]

75. [16 marks] Let $C = \begin{pmatrix} 2 & 1 \\ 1 & 2 \\ 1 & -1 \end{pmatrix}$.

(a) Compute $C^{\mathrm{T}}C$, and find its eigenvalues and a unit eigenvector for each. [4]

(b) Write down the singular values of C and the matrices Σ and V. [3]

(c) Find an orthogonal matrix $\boldsymbol{U}$ such that $\boldsymbol{C} = \boldsymbol{U}\Sigma\boldsymbol{V}^{\mathrm{T}}$. [5]

(d) Use your decomposition to write down an orthonormal basis for $\mathrm{Col}(\boldsymbol{C})$ and one for $\mathrm{Null}\left(\boldsymbol{C}^{\mathrm{T}}\right)$, and state $\mathrm{rank}(\boldsymbol{C})$. [4]

Answers

Final answers with the key step. A written answer in the examination needs the full working.

Part A

1. RREF $\left(\begin{array}{cccc|c} 1 & 2 & 0 & 3 & 5 \\ 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$. Columns 2 and 4 are non-pivot, so $x_2 = s$, $x_4 = t$: $\boldsymbol{x} = (5 - 2s - 3t, s, 2 - 2t, t)^{\mathrm{T}}$.

2. $\det = -(a-1)^2(a+2)$. (i) $a = 1$ (the equations read $x + y + z = 1$ and $x + y + z = -2$). (ii) $a \neq 1, -2$. (iii) $a = -2$: $(x, y, z) = (1 + s, s, s)$.

3. $R_2 - 2R_1$ and $R_3 + R_1$ give rows $(0, c-4, 0, c-4)$ and $(0, 0, d-1, d-1)$. Rank 3 if $c \neq 4$ and $d \neq 1$; rank 2 if exactly one of $c = 4$, $d = 1$ holds; rank 1 if $c = 4$ and $d = 1$. Nullity $= 4 -$ rank: 1, 2, 3 respectively.

4. (a) $\boldsymbol{A}^{-1} = \begin{pmatrix} -2 & -5 & 3 \\ 2 & 3 & -2 \\ -1 & -1 & 1 \end{pmatrix}$. (b) $\boldsymbol{X} = \boldsymbol{A}^{-1}\boldsymbol{B} = \begin{pmatrix} -12 & 4 \\ 8 & -3 \\ -3 & 2 \end{pmatrix}$.

5. (a) $\boldsymbol{E}_1 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$, $\boldsymbol{E}_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, $\boldsymbol{E}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}$. (b) $\det(\boldsymbol{B}) = (-1)(3)(1)(4) = -12$; $\det\left(2\boldsymbol{A}^{-1}\boldsymbol{B}^{\mathrm{T}}\right) = 2^3 \cdot \frac{1}{4} \cdot (-12) = -24$.

6. (a) $\boldsymbol{L} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 2 & 1 \end{pmatrix}$, $\boldsymbol{U} = \begin{pmatrix} 2 & 1 & -1 \\ 0 & 3 & -1 \\ 0 & 0 & 4 \end{pmatrix}$. (b) $\boldsymbol{y} = (1, 5, 4)^{\mathrm{T}}$, $\boldsymbol{x} = (0, 2, 1)^{\mathrm{T}}$.

7. (a) 2, 3, 4. (b) The $(1,1)$-cofactor is d_{n-1}; the $(1,2)$-minor has first column $(1, 0, \ldots, 0)^{\mathrm{T}}$, and expanding along it gives d_{n-2}. (c) $d_n - d_{n-1} = d_{n-1} - d_{n-2} = 1$, so $d_n = n + 1 \neq 0$: invertible for every n.

8. $\det(\boldsymbol{M}) = 1 - 2t^2$; nontrivial solutions exactly when $t = \pm 1/\sqrt{2}$.

9. 8; 27/2; $\det(\boldsymbol{A})^{n-1} = 4$; $\det\left(\boldsymbol{A}^{\mathrm{T}}\right)\det(\boldsymbol{A} - 2\boldsymbol{I}) = 10$.

10. (a) False: $\boldsymbol{A} = \boldsymbol{B} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. (b) False: it equals $\boldsymbol{A}^2 + \boldsymbol{A}\boldsymbol{B} + \boldsymbol{B}\boldsymbol{A} + \boldsymbol{B}^2$; take $\boldsymbol{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $\boldsymbol{B} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. (c) False: $\boldsymbol{A} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. (d) False: $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. (e) True: premultiply by $\boldsymbol{A}^{-1}$.

Part B

11. $\boldsymbol{v} \in \operatorname{span}(S)$ with $[\boldsymbol{v}]_S = (2, 0, 1)^{\mathrm{T}}$. $\boldsymbol{w} \notin \operatorname{span}(S)$: the last column of the augmented matrix is a pivot column.

12. (a) Subspace, $\operatorname{span}\{(2,1,0)^{\mathrm{T}}, (-1,0,1)^{\mathrm{T}}\}$. (b) No: $\boldsymbol{0}$ is not in it. (c) Subspace, $\operatorname{span}\{(1,1,2)^{\mathrm{T}}, (1,-1,0)^{\mathrm{T}}\}$. (d) No: $(1,0)^{\mathrm{T}}$ and $(0,-1)^{\mathrm{T}}$ are in it, their sum $(1,-1)^{\mathrm{T}}$ is not. (e) Subspace: it is $\{x = y = 0\} = \operatorname{span}\{(0,0,1)^{\mathrm{T}}\}$.

13. $\det = 1 + k^3 = (k+1)(k^2 - k + 1)$, so only $k = -1$. Then the three vectors sum to $\boldsymbol{0}$: $(-1,0,1)^{\mathrm{T}} = -(1,-1,0)^{\mathrm{T}} - (0,1,-1)^{\mathrm{T}}$.

14. (a) $\{(-3,-1,1,0)^{\mathrm{T}}, (1,1,0,1)^{\mathrm{T}}\}$, $\dim V = 2$. (b) Add $\boldsymbol{e}_1, \boldsymbol{e}_2$: in a vanishing combination, rows 3 and 4 force the first two coefficients to be 0, then $\boldsymbol{e}_1, \boldsymbol{e}_2$ are independent. (Other answers are possible.)

15. (a) $\{\boldsymbol{u}_1, \boldsymbol{u}_3, \boldsymbol{u}_5\}$ (pivot columns 1, 3, 5). (b) $\boldsymbol{u}_2 = 2\boldsymbol{u}_1$, $\boldsymbol{u}_4 = \boldsymbol{u}_1 + \boldsymbol{u}_3$.

16. (a) $\boldsymbol{P} = \frac{1}{2}\begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 1 \\ 1 & 0 & -1 \end{pmatrix}$ (columns are $[\boldsymbol{v}_i]_S$). (b) $[\boldsymbol{w}]_S = (0, 2, 1)^{\mathrm{T}}$, $\boldsymbol{w} = (2, 3, 1)^{\mathrm{T}}$. (c) $\boldsymbol{P}^{-1} = \begin{pmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \\ 1 & 0 & -1 \end{pmatrix}$.

17. $\dim(U + V)$ is 4 or 5, so $\dim(U \cap V)$ is 3 or 2. If $U \not\subseteq V$ then $U + V \neq V$, so $\dim(U + V) = 5$ and $\dim(U \cap V) = 2$.

18. (a) $a(\boldsymbol{u} + \boldsymbol{v}) + b(\boldsymbol{v} + \boldsymbol{w}) + c(\boldsymbol{w} + \boldsymbol{u}) = \boldsymbol{0}$ gives $a + c = a + b = b + c = 0$, so $a = b = c = 0$. (b) The three vectors sum to $\boldsymbol{0}$.

Part C

19. (a) $\begin{pmatrix} 1 & 2 & 0 & 1 & -1 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$. (b) Row space: its two nonzero rows. Column space: columns 1 and 3 of $\boldsymbol{A}$, $(1, 2, -1, 3)^{\mathrm{T}}$, $(0, 1, 1, 1)^{\mathrm{T}}$. Nullspace: $(-2, 1, 0, 0, 0)^{\mathrm{T}}$, $(-1, 0, -2, 1, 0)^{\mathrm{T}}$, $(1, 0, -1, 0, 1)^{\mathrm{T}}$. (c) 2, 3, $4 - 2 = 2$.

20. (a) Nullity 2, rank 3; basis $(1, 1, 0, 0, 0)^{\mathrm{T}}$, $(-2, 0, 1, 1, 0)^{\mathrm{T}}$. (b) Yes, since the rank equals the number of rows; so there is a right inverse. No left inverse: the rank is less than 5.

21. (a) $\operatorname{rank}(\boldsymbol{A}) = 2$ is the number of rows but not the number of columns. (b) Solving $\boldsymbol{A}\boldsymbol{X} = \boldsymbol{I}$ column by column: $\boldsymbol{X} = \begin{pmatrix} 1 - 2s & -2t \\ s & 1 + t \\ s & t \end{pmatrix}$, $s, t \in \mathbb{R}$.

22. $R_3 - 2R_1 - R_2$ gives the zero row, so the system is consistent exactly when $b_3 = 2b_1 + b_2$: $\operatorname{Col}(\boldsymbol{A}) = \{\boldsymbol{b} : 2b_1 + b_2 - b_3 = 0\}$.

23. (a) True. (b) False: $\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$ and $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$. (c) True: $\operatorname{Col}(\boldsymbol{A}\boldsymbol{B}) \subseteq \operatorname{Col}(\boldsymbol{A})$. (d) False: nullity $\geq 7 - 5 = 2$. (e) True: $\operatorname{Col}(\boldsymbol{A}) = \mathbb{R}^m$.

24. (a) $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}$ gives $\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}$. Conversely $\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}$ gives $\|\boldsymbol{A}\boldsymbol{x}\|^2 = \boldsymbol{x}^{\mathrm{T}}\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}\boldsymbol{x} = 0$, so $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}$. (b) Both matrices have n columns and the same nullity, hence the same rank; $\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}$ is $n \times n$, so it is invertible iff $\operatorname{rank}(\boldsymbol{A}) = n$.

Part D

25. (a) All three dot products are 0; an orthogonal set of nonzero vectors is linearly independent. (b) $(\boldsymbol{w} \cdot \boldsymbol{u}_i)/4$ gives $[\boldsymbol{w}] = (1, 1, 2)^{\mathrm{T}}$, and $\boldsymbol{u}_1 + \boldsymbol{u}_2 + 2\boldsymbol{u}_3 = \boldsymbol{w}$. (c) $\frac{1}{4}(\boldsymbol{u}_1 + \boldsymbol{u}_2 + \boldsymbol{u}_3) = \left(\frac{3}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}\right)^{\mathrm{T}}$; distance $\frac{1}{2}$.

26. $\{(2, -2, 1, 0)^{\mathrm{T}}, (2, -1, 0, 1)^{\mathrm{T}}\}$ (the nullspace of the matrix with the two given rows); $2 + 2 = 4$.

27. (a) $\boldsymbol{P} = \frac{1}{9}\begin{pmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}$. (b) $\boldsymbol{I} - \boldsymbol{P} = \frac{1}{9}\begin{pmatrix} 8 & -2 & -2 \\ -2 & 5 & -4 \\ -2 & -4 & 5 \end{pmatrix}$. (c) Eigenvalue 1 with eigenspace V; eigenvalue 0 with eigenspace $V^{\perp}$, the plane $x + 2y + 2z = 0$.

28. $\boldsymbol{v}_1 = (1, 1, 0, 0)^{\mathrm{T}}$, $\boldsymbol{v}_2 = \left(\frac{1}{2}, -\frac{1}{2}, 1, 0\right)^{\mathrm{T}}$, $\boldsymbol{v}_3 = \left(-\frac{2}{3}, \frac{2}{3}, \frac{2}{3}, 1\right)^{\mathrm{T}}$. Normalized: $\frac{1}{\sqrt{2}}(1, 1, 0, 0)^{\mathrm{T}}$, $\frac{1}{\sqrt{6}}(1, -1, 2, 0)^{\mathrm{T}}$, $\frac{1}{\sqrt{21}}(-2, 2, 2, 3)^{\mathrm{T}}$.

29. (a) $\boldsymbol{Q} = \frac{1}{3}\begin{pmatrix} 1 & -2 \\ 2 & 2 \\ 2 & -1 \end{pmatrix}$, $\boldsymbol{R} = \begin{pmatrix} 3 & 3 \\ 0 & 3 \end{pmatrix}$. (b) $\boldsymbol{R}\hat{\boldsymbol{x}} = \boldsymbol{Q}^{\mathrm{T}}\boldsymbol{b} = (3, -3)^{\mathrm{T}}$, so $\hat{\boldsymbol{x}} = (2, -1)^{\mathrm{T}}$.

30. (a) $Q^{\mathrm{T}}Q = I$; $\det(Q) = -1$. (b) $\boldsymbol{x} = Q^{\mathrm{T}}\boldsymbol{b} = (10, 5, -1)^{\mathrm{T}}$.

31. $A^{\mathrm{T}}A = \begin{pmatrix} 4 & 2 \\ 2 & 6 \end{pmatrix}$, $A^{\mathrm{T}}\boldsymbol{b} = (8, 11)^{\mathrm{T}}$: $y = 1.3 + 1.4x$.

32. (a) $R_1 - R_2 - R_3$ gives $0 = -4$. (b) $A^{\mathrm{T}}A = \begin{pmatrix} 2 & 1 & 3 \\ 1 & 2 & 3 \\ 3 & 3 & 6 \end{pmatrix}$, $A^{\mathrm{T}}\boldsymbol{b} = (3, 4, 7)^{\mathrm{T}}$: $\hat{\boldsymbol{x}} = \left(\frac{2}{3} - s, \frac{5}{3} - s, s\right)^{\mathrm{T}}$. (c) $\left(\frac{7}{3}, \frac{2}{3}, \frac{5}{3}\right)^{\mathrm{T}}$. Any two least squares solutions differ by a vector of $\mathrm{Null}\left(A^{\mathrm{T}}A\right) = \mathrm{Null}(A)$, so $A\hat{\boldsymbol{x}}$ is the same.

33. (a) $(Q\boldsymbol{x}) \cdot (Q\boldsymbol{y}) = \boldsymbol{x}^{\mathrm{T}}Q^{\mathrm{T}}Q\boldsymbol{y} = \boldsymbol{x}^{\mathrm{T}}\boldsymbol{y}$; take $\boldsymbol{y} = \boldsymbol{x}$ for the norm. (b) $QQ^{\mathrm{T}}\boldsymbol{b} \in \mathrm{Col}(Q)$ and $Q^{\mathrm{T}}\left(\boldsymbol{b} - QQ^{\mathrm{T}}\boldsymbol{b}\right) = \boldsymbol{0}$, so the remainder is orthogonal to $\mathrm{Col}(Q)$.

Part E

34. (a) $(x-1)(x-2)^2$; eigenvalues 1, 2. (b) $E_1 = \mathrm{span}\left\{(1, 0, 1)^{\mathrm{T}}\right\}$ (multiplicities 1, 1); $E_2 = \mathrm{span}\left\{(1, 1, 0)^{\mathrm{T}}, (0, 0, 1)^{\mathrm{T}}\right\}$ (multiplicities 2, 2). (c) $P = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$, $D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$.

35. The eigenvalue 1 has algebraic multiplicity 2, and $C - I = \begin{pmatrix} 0 & a & b \\ 0 & 1 & c \\ 0 & 0 & 0 \end{pmatrix}$ has rank 1 exactly when $b = ac$. So C is diagonalizable iff $b = ac$.

36. (a) $P = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}$, $D = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$. (b) $A^n = \begin{pmatrix} 2 \cdot 3^n - 2^n & 2^{n+1} - 2 \cdot 3^n \\ 3^n - 2^n & 2^{n+1} - 3^n \end{pmatrix}$.

37. $\det = -6$; trace 2; A^{-1}: $1, -\frac{1}{2}, \frac{1}{3}$; $A^2 + I$: $2, 5, 10$; $\det\left(A^2 - A\right) = \prod \lambda(\lambda - 1) = 0$; $A + 2I$ is singular because -2 is an eigenvalue.

38. Eigenvalues 5 and 2 (twice). $P = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & -\frac{2}{\sqrt{6}} \end{pmatrix}$, $D = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$.

39. (a) A is diagonalizable, so $\dim E_4 = 2$, and $E_4 \perp E_1$: E_4 is the plane $x + y + z = 0$. (b) With J the all-ones matrix, $A = \frac{1}{3}J + 4\left(I - \frac{1}{3}J\right) = 4I - J = \begin{pmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{pmatrix}$.

40. (a) $P = \begin{pmatrix} 0.5 & 0.2 & 0.3 \\ 0.3 & 0.6 & 0.3 \\ 0.2 & 0.2 & 0.4 \end{pmatrix}$. (b) $P^2\boldsymbol{e}_1 = (0.37, 0.39, 0.24)^{\mathrm{T}}$: 0.39. (c) $\left(\frac{9}{28}, \frac{12}{28}, \frac{7}{28}\right)^{\mathrm{T}}$: in the long run about 32%, 43% and 25% of the bikes are at X, Y, Z.

41. (a) $P = \begin{pmatrix} 4 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}\begin{pmatrix} 4 & 1 \\ 1 & -1 \end{pmatrix}^{-1}$. (b) $\boldsymbol{x}_n = (0.8 + 0.2 \cdot 2^{-n}, 0.2 - 0.2 \cdot 2^{-n})^{\mathrm{T}} \to (0.8, 0.2)^{\mathrm{T}}$.

42. $A^{\mathrm{T}}A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$, so $\sigma_1 = \sqrt{3}$, $\sigma_2 = \sqrt{2}$, $\boldsymbol{v}_1 = \boldsymbol{e}_2$, $\boldsymbol{v}_2 = \boldsymbol{e}_1$.

$$U = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \end{pmatrix}, \Sigma = \begin{pmatrix} \sqrt{3} & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{pmatrix}, V = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

43. $AA^{\mathrm{T}} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$: $\sigma_1 = \sqrt{3}$, $\sigma_2 = 1$.

$$U = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \Sigma = \begin{pmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, V = \begin{pmatrix} \frac{2}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} \end{pmatrix}.$$

44. (a) Rank 2; eigenvalues $144, 36, 0$. (b) $\boldsymbol{x} = s(2, -2, 1)^{\mathrm{T}}$. (c) $(\boldsymbol{w} \cdot \boldsymbol{u}_1)\boldsymbol{u}_1 + (\boldsymbol{w} \cdot \boldsymbol{u}_2)\boldsymbol{u}_2 = 5\boldsymbol{u}_1 - \boldsymbol{u}_2 = (2, 3, 2, 3)^{\mathrm{T}}$. (d) $A = 12\boldsymbol{u}_1\boldsymbol{v}_1^{\mathrm{T}} + 6\boldsymbol{u}_2\boldsymbol{v}_2^{\mathrm{T}} = \begin{pmatrix} 4 & 5 & 2 \\ 0 & 3 & 6 \\ 4 & 5 & 2 \\ 0 & 3 & 6 \end{pmatrix}$.

45. (a) False: $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. (b) False: the zero matrix. (c) True. (d) True: $A^{\mathrm{T}} = \left(P^{\mathrm{T}}\right)^{-1}DP^{\mathrm{T}}$. (e) False: I and $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

46. (a) If $A\boldsymbol{v} = \lambda\boldsymbol{v}$ with $\boldsymbol{v} \neq \boldsymbol{0}$, then $\lambda\boldsymbol{v} = A\boldsymbol{v} = A^2\boldsymbol{v} = \lambda^2\boldsymbol{v}$, so $\lambda^2 = \lambda$. (b) Let $AB\boldsymbol{v} = \lambda\boldsymbol{v}$, $\boldsymbol{v} \neq \boldsymbol{0}$, and $\boldsymbol{w} = B\boldsymbol{v}$. Then $A\boldsymbol{w} = \lambda\boldsymbol{v} \neq \boldsymbol{0}$ so $\boldsymbol{w} \neq \boldsymbol{0}$, and $BA\boldsymbol{w} = B(\lambda\boldsymbol{v}) = \lambda\boldsymbol{w}$.

Part F

47. (a) Linear, $\begin{pmatrix} 1 & 2 \\ 0 & 3 \\ 1 & 0 \end{pmatrix}$. (b) Not linear: $T(\boldsymbol{0}) \neq \boldsymbol{0}$. (c) Not linear: $T(2, 2) = (4, 2) \neq 2T(1, 1)$. (d) Linear, $\boldsymbol{a}\boldsymbol{a}^{\mathrm{T}} = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 0 & 0 \\ 2 & 0 & 4 \end{pmatrix}$.

48. (a) $[T]_S = \begin{pmatrix} 1 & 0 & 3 \\ 2 & 1 & -1 \end{pmatrix}$. (b) $A = [T]_S P$ with $P = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}^{-1}$: $A = \begin{pmatrix} 2 & -1 & 1 \\ 0 & 2 & -1 \end{pmatrix}$, so $T(x, y, z) = (2x - y + z, 2y - z)$.

49. (a) $[T]_S = \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}$. (b) $A = [T]_S\begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}^{-1} = \frac{1}{5}\begin{pmatrix} -3 & 4 \\ 4 & 3 \end{pmatrix}$; $A^2 = I$, $\det(A) = -1$.

50. $Q \circ R$: $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix}$; $R \circ Q$: $\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. Kernel and range of $Q \circ R$ are both $\mathrm{span}\{\boldsymbol{e}_1\}$.

51. RREF of the standard matrix: $\begin{pmatrix} 1 & 0 & -2 & -3 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$. (a) Kernel: $(2, -1, 1, 0)^{\mathrm{T}}$, $(3, -1, 0, 1)^{\mathrm{T}}$; range: $(1, 0, 1)^{\mathrm{T}}$, $(2, 1, 3)^{\mathrm{T}}$. (b) Rank 2, nullity 2; neither injective nor surjective.

52. (a) $\det = a\left(a^2 - 2\right)$: $a = 0, \pm\sqrt{2}$. (b) $a = 0$: $\ker(T) = \mathrm{span}\left\{(1, 0, -1)^{\mathrm{T}}\right\}$.

53. (a) 1: $(1, 0, 0)^{\mathrm{T}}$; $\frac{1}{2}$: $(-2, 1, 0)^{\mathrm{T}}$; $\frac{1}{3}$: $(9, -6, 1)^{\mathrm{T}}$. (b) Writing $\boldsymbol{x}$ in this eigenbasis, the coefficient of $(1, 0, 0)^{\mathrm{T}}$ is $x_1 + 2x_2 + 3x_3$ and the other two terms decay, so the limit is $(x_1 + 2x_2 + 3x_3, 0, 0)^{\mathrm{T}}$.

54. (a) If T is injective and $T(\boldsymbol{x}) = \boldsymbol{0} = T(\boldsymbol{0})$ then $\boldsymbol{x} = \boldsymbol{0}$. Conversely $T(\boldsymbol{x}_1) = T(\boldsymbol{x}_2)$ gives $\boldsymbol{x}_1 - \boldsymbol{x}_2 \in \ker(T) = \{\boldsymbol{0}\}$. (b) The standard matrix is $m \times n$: nullity $\geq n - m > 0$ if $n > m$; rank $\leq n < m$ if $n < m$.

Part G Section A

55. **A.** (II) has a leading entry 2; (III) has a zero row above a nonzero row.

56. **B.** More variables than equations; the number of parameters is $6 - \mathrm{rank} \geq 2$.

57. **C.** $\det(B) = -10$, so $\frac{1}{5} \cdot (-10) = -2$.

58. **C.** (I) fails for $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$; (II) by $\det(A)\det(B) \neq 0$.

59. **D.** Only (II) is a subspace, which none of (A)–(C) says.

60. **C.** The vectors in (I) sum to $\boldsymbol{0}$.

61. **D.** $7-3=4$ and $4-3=1$.

62. **C.** Row operations preserve the row space and the nullspace only.

63. **A.** Orthogonal nonzero vectors are independent, and at most n vectors in $\mathbb{R}^n$ are.

64. **A.** Unique only when the columns of A are independent; $A\hat{x}$ is always the projection.

65. **B.** $\frac{6}{3}(1,1,1)^{\mathrm{T}}$.

66. **D.** (I) has distinct eigenvalues, (III) is symmetric; (II) has a one-dimensional eigenspace.

67. **A.** 0 is an eigenvalue; $\begin{pmatrix}0&1&0\\0&0&0\\0&0&3\end{pmatrix}$ is not diagonalizable and has rank 2.

68. **D.** $0.3x_1 = 0.2x_2$ with $x_1 + x_2 = 1$.

69. **B.** The eigenvalues of $A^{\mathrm{T}}A$ are 9 and 1.

70. **B.** Nullity ≥ 2; the zero map is not surjective.

Part G Section B

71. (a) $\det(A) = a(2-a)$. (b) $a \neq 0, 2$. (c) $A^{-1} = \begin{pmatrix}3&-1&-1\\-1&1&0\\-1&0&1\end{pmatrix}$. (d) Rows 2 and 3 of A coincide, so $c = 3$; then $x = (-1, 2-s, s)^{\mathrm{T}}$.

72. (a) Each u_i satisfies the equation, S is linearly independent (solve from the first coordinate down), and $\dim V = 4-1=3$. (b) $2-3+4-3=0$; $[w]_S = (2,1,3)^{\mathrm{T}}$. (c) With $n = (1,-1,1,-1)^{\mathrm{T}}$: $y - \frac{y\cdot n}{n\cdot n}n = y - n = (2,0,0,2)^{\mathrm{T}}$. (d) $I - \frac{1}{4}nn^{\mathrm{T}} = \frac{1}{4}\begin{pmatrix}3&1&-1&1\\1&3&1&-1\\-1&1&3&1\\1&-1&1&3\end{pmatrix}$. (e) $\ker(T) = \operatorname{span}\{n\}$, range V, rank 3, nullity 1.

73. (a) $Q = \frac{1}{2}\begin{pmatrix}1&1\\1&-1\\1&1\\1&-1\end{pmatrix}$, $R = \begin{pmatrix}2&4\\0&2\end{pmatrix}$. (b) $Q^{\mathrm{T}}b = (4,4)^{\mathrm{T}}$, $\hat{x} = (-2,2)^{\mathrm{T}}$. (c) $A\hat{x} = (4,0,4,0)^{\mathrm{T}}$; distance $\left\|(-1,1,1,-1)^{\mathrm{T}}\right\| = 2$.

74. (a) $\lambda = 2$: $\operatorname{span}\{(2,1)^{\mathrm{T}}\}$; $\lambda = -1$: $\operatorname{span}\{(1,-1)^{\mathrm{T}}\}$. (b) $P = \begin{pmatrix}2&1\\1&-1\end{pmatrix}$, $D = \begin{pmatrix}2&0\\0&-1\end{pmatrix}$. (c) $A^n = \frac{1}{3}\begin{pmatrix}2^{n+1}+(-1)^n & 2^{n+1}-2(-1)^n\\ 2^n-(-1)^n & 2^n+2(-1)^n\end{pmatrix}$. (d) $x_n = \frac{1}{3}\left(2^{n+1}+(-1)^n,\ 2^n-(-1)^n\right)^{\mathrm{T}}$; the ratio tends to 2.

75. (a) $C^{\mathrm{T}}C = \begin{pmatrix}6&3\\3&6\end{pmatrix}$: 9 with $\frac{1}{\sqrt{2}}(1,1)^{\mathrm{T}}$, 3 with $\frac{1}{\sqrt{2}}(1,-1)^{\mathrm{T}}$. (b) $\sigma_1 = 3$, $\sigma_2 = \sqrt{3}$; $\Sigma = \begin{pmatrix}3&0\\0&\sqrt{3}\\0&0\end{pmatrix}$, $V = \frac{1}{\sqrt{2}}\begin{pmatrix}1&1\\1&-1\end{pmatrix}$. (c) $u_1 = \frac{1}{3}Cv_1 = \frac{1}{\sqrt{2}}(1,1,0)^{\mathrm{T}}$, $u_2 = \frac{1}{\sqrt{3}}Cv_2 = \frac{1}{\sqrt{6}}(1,-1,2)^{\mathrm{T}}$, $u_3 = \frac{1}{\sqrt{3}}(1,-1,-1)^{\mathrm{T}}$; $U = (u_1\ u_2\ u_3)$. (d) $\operatorname{Col}(C)$: $\{u_1, u_2\}$; $\operatorname{Null}(C^{\mathrm{T}})$: $\{u_3\}$; rank 2.

Scoring the mock paper: 2 marks for each correct letter in Section A (32), and the marks shown for each part of Section B (68). Signs of singular vectors and the choice of eigenvectors may differ from those above.